

CSE 150A-250A AI: Probabilistic Methods

Lecture 3

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Joint Distributions and Inference

Alarm Example

Belief networks

Review

- Types of probabilities:

$P(X, Y)$ joint

$P(Y|X)$ conditional

$P(X)$ unconditional (or marginal)

- Useful rules:

$$P(A, B, C, \dots) = P(A) P(B|A) P(C|A, B) \dots \quad \text{product rule}$$

$$P(X|Y) = P(Y|X)P(X)/P(Y) \quad \text{Bayes rule}$$

$$P(X) = \sum_y P(X, Y=y) \quad \text{marginalization}$$

- Conditioning on background evidence E :

$$P(A, B, C, \dots | E) = P(A|E) P(B|A, E) P(C|A, B, E) \dots$$

$$P(X|Y, E) = P(Y|X, E)P(X|E)/P(Y|E)$$

$$P(X|E) = \sum_y P(X, Y=y|E)$$

Marginal and conditional independence

- Marginal independence

$$P(X|Y) = P(X)$$

$$P(Y|X) = P(Y)$$

$$P(X, Y) = P(X) P(Y)$$

*Each of these
implies the
other two.*

- Conditional independence

$$P(X|Y, E) = P(X|E)$$

$$P(Y|X, E) = P(Y|E)$$

$$P(X, Y|E) = P(X|E) P(Y|E)$$

*Each of these
also implies the
other two.*

Example of conditional dependence



- B and E are marginally independent:

$$P(B) = P(B|E)$$

$$P(E) = P(E|B)$$

$$P(B, E) = P(B)P(E)$$

- But B and E are conditionally dependent given A :

$$P(B|A) \neq P(B|E, A)$$

$$P(E|A) \neq P(E|B, A)$$

$$P(B, E|A) \neq P(B|A)P(E|A)$$

Joint Distributions and Inference

Why Joints Matter

The joint distribution $P(X_1, \dots, X_n)$ is a **complete description of uncertainty**.

From the full joint distribution $P(X_1, \dots, X_n)$, what kinds of probabilities can we compute?

- A. Only marginals, $P(X_i)$
- B. Only conditionals, $P(X_i|X_j)$
- C. Any marginal or conditional over the variables
- D. None of the above

Inference: Compute the **posterior** distribution of query variables given observed evidence.

Motivation

- Model complexity

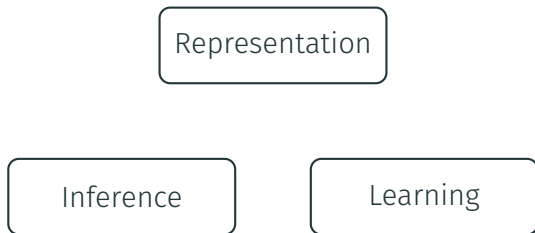
Suppose $X_i \in \{0, 1\}$ are binary random variables.

How many numbers do we need to specify the joint distribution of $P(X_1=x_1, \dots, X_n=x_n)$?

- A. $O(n)$
- B. $O(2^n)$
- C. $O(n^2)$
- D. $O(\log n)$

It requires numbers to specify the joint distribution $P(X_1=x_1, \dots, X_n=x_n)$.

Conceptual and Practical Goals



- **Representation:** compactly encode the joint.
- **Inference:** answer queries given evidence.
- **Learning:** estimate structure/parameters from data.

Alarm Example

Alarm example



- Binary random variables

$B \in \{0, 1\}$ Was there a burglary?

$E \in \{0, 1\}$ Was there an earthquake?

$A \in \{0, 1\}$ Was the alarm triggered?

$J \in \{0, 1\}$ Did Jamal call?

$M \in \{0, 1\}$ Did Maya call?

- Product rule

$$\begin{aligned} P(B, E, A, J, M) \\ = P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J) \end{aligned}$$

Note: the above is true no matter what the variables signify.

- Domain-specific assumptions

$P(E B)$	$=$	$P(E)$	marginal independence
$P(J B, E, A)$	$=$	$P(J A)$	conditional independence
$P(M B, E, A, J)$	$=$	$P(M A)$	conditional independence

Completing the model

- Joint distribution

$$P(B, E, A, J, M)$$

$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

$$= P(B) P(E) P(A|B, E) P(J|A) P(M|A)$$

- Conditional probability tables (CPTs)

$P(B=1) = 0.001$

$P(E=1) = 0.002$

B	E	$P(A=1 B, E)$
0	0	0.001
1	0	0.94
0	1	0.29
1	1	0.95

A	$P(J=1 A)$
0	0.05
1	0.9

A	$P(M=1 A)$
0	0.01
1	0.7

- Joint probabilities are easy to compute:

$$\begin{aligned} &P(B=1, E=0, A=1, J=1, M=1) \\ &= P(B=1) P(E=0) P(A=1|B=1, E=0) P(J=1|A=1) P(M=1|A=1) \\ &= (0.001)(1 - 0.002)(0.94)(0.9)(0.7) \end{aligned}$$

- Any inference can be expressed in terms of joint probabilities:

$$\begin{aligned} &P(B=1, E=0|M=1) \\ &= \frac{P(B=1, E=0, M=1)}{P(M=1)} \quad \boxed{\text{product rule}} \\ &= \frac{\sum_{a,j} P(B=1, E=0, A=a, J=j, M=1)}{\sum_{b',e',a',j'} P(B=b', E=e', A=a', J=j', M=1)} \quad \boxed{\text{marginalization}} \end{aligned}$$

But this approach can be very inefficient!

How to perform inference more efficiently?

1. Visualize models as directed acyclic graphs.
2. Exploit graph structure to organize and simplify calculations.

Belief networks

A **belief network** (BN) is a directed acyclic graph (DAG) in which:

1. Nodes represent random variables.
2. Edges represent direct dependencies in the factorization.
3. Conditional probability tables (CPTs) describe how each node depends on its parents.

$$\text{BN} = \text{DAG} + \text{CPTs}$$

From distributions to graphs

- It is always true from the product rule that

$$\begin{aligned} P(X_1, X_2, \dots, X_n) &= P(X_1) P(X_2|X_1) \dots P(X_n|X_1, \dots, X_{n-1}) \\ &= \prod_{i=1}^n P(X_i|X_1, X_2, \dots, X_{i-1}) \end{aligned}$$

- But suppose in a particular domain that

$$P(X_i|X_1, X_2, \dots, X_{i-1}) = P(X_i|\text{parents}(X_i)),$$

where $\text{parents}(X_i)$ is a subset of $\{X_1, \dots, X_{i-1}\}$.

- **Big idea:** represent conditional dependencies by a DAG.

Visualizing the model - Alarm Example

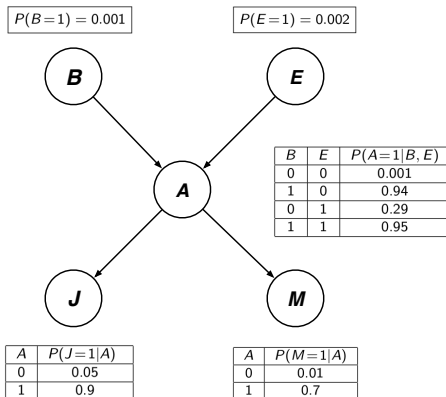
- Joint distribution

$$P(B, E, A, J, M)$$

$$= P(B) P(E|B) P(A|B, E) P(J|B, E, A) P(M|B, E, A, J)$$

$$= P(B) P(E) P(A|B, E) P(J|A) P(M|A)$$

Alarm belief network



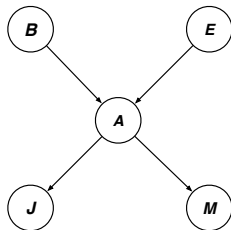
This visual representation of the joint distribution is called a **belief network** (or a **Bayesian network**, or a **probabilistic graphical model**).

- Given a BN, the graph and CPTs specify the joint distribution.

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{parents}(X_i))$$

- To read the factorization from a graph:
 1. write one factor for each node;
 2. condition each node on its parents;
 3. multiply the factors.

Alarm BN:



$$P(B, E, A, J, M) = P(B) P(E) P(A|B, E) P(J|A) P(M|A)$$

Example: graph $\rightarrow$ factorization $\rightarrow$ independence



What independence relationship is implied by this BN?

- General product rule:

$$P(X, Y, Z) = P(X) P(Y|X) P(Z|X, Y).$$

- BN factorization:

$$P(X, Y, Z) = P(X) P(Y|X) P(Z|Y).$$

- Comparing these factorizations gives

$$P(Z|X, Y) = P(Z|Y).$$

$$X \perp Z \mid Y$$

Advantages of belief networks

1. Compact representation of complex models

BNs provide a complete but parsimonious representation of joint probability distributions.

DAG + CPTs $\Rightarrow$ joint distribution

2. Crisp separation of qualitative vs quantitative knowledge

Qualitative

DAGs encode assumptions of marginal and conditional independence.

Quantitative

CPTs encode numerical influences of some variables on others.

That's all folks!

Addendum

Constructing BNs and node ordering

Constructing a belief network

Three steps:

1. Choose your random variables of interest.
2. Choose an ordering of these variables (e.g., $X_1, X_2, \dots, X_n$).
3. While there are variables left:
 - (a) add the node X_i to the network
 - (b) set the parents of X_i to be the minimal subset satisfying

$$P(X_1, X_2, \dots, X_n) = \prod_{i=1}^n P(X_i | \text{parents}(X_i)),$$

- (c) define the conditional probability table $P(X_i | \text{parents}(X_i))$

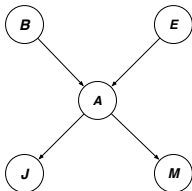
Node ordering

- **Common heuristic:**

Add the “root causes,” then the variables they influence, then the next variables that are influenced, etc. This often gives a compact graph, but it is not a formal guarantee of optimality.

- **Example:**

In the alarm world, a natural ordering is (B, E, A, J, M) .



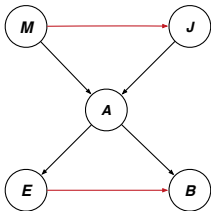
Node ordering

- What happens if we choose an unnatural ordering?

Ex: (M, J, A, E, B)

- Adding nodes with this ordering:

$$\begin{aligned} P(M, J, A, E, B) \\ &= P(M) P(J|M) P(A|J, M) P(E|A, J, M) P(B|E, A, J, M) \\ &= P(M) P(J|M) P(A|J, M) P(E|A) P(B|A, E) \end{aligned}$$



This belief network has **two extra edges**.
This DAG does not show $P(B) = P(B|E)$.
This DAG does not show $P(M|A) = P(M|A, J)$.
This belief network has **larger CPTs**.
These CPTs may be more difficult to assess.